

vMF Hypersphere Music Representation

Visualizing harmony, chord quality, and root relationships as directions on a learned vMF hypersphere

Pearson
0.954

Fifths distance vs vMF angle

Spearman
0.953

Strong rank correspondence

mu dim
8

Learned direction space

Root
12 classes

C to B pitch classes

Core idea

Instead of treating harmony only as discrete labels, this project represents musical states as directions on a von Mises-Fisher hypersphere.

What was verified

Root pairs close on the circle of fifths also tend to be close in learned vMF angular distance. The direction space partially reflects tonal proximity.

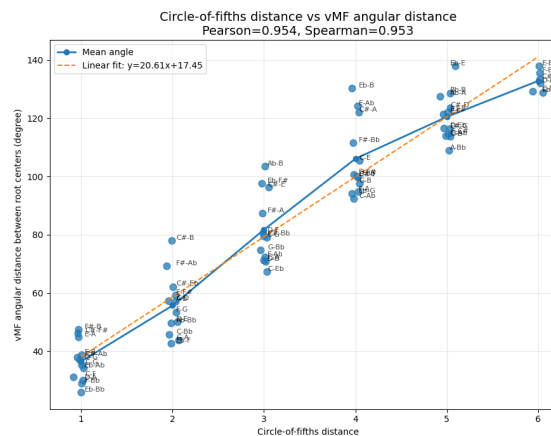
Connection to performance generation

The harmonic center μ_t can be compared with each note direction e_{ti} . The alignment score $A_{ti} = e_{ti}^T \mu_t$ can support velocity and accompaniment-expression control.

Scope and limitations

Root labels are inferred from pitch-class sets. 3D plots are PCA projections from the original 8D mu space. Copyrighted MIDI files, private datasets, and large checkpoints are not included.

Circle of fifths vs vMF angle



Chord-type angular distance

